



May 2026

Research Ethics Application Checklist

All sections of the form should be completed. You must append, at the end of this form, all of the materials for your project (including a consent for participation). The staff member reviewing this form needs to see full details of what your participants will experience (including access to archive materials). In completing this form, you must follow the guidance provided by your subject area. Some important information to note.

- This form shall be submitted via the research ethics committee ethics application form.
- The application shall be made one month in advance before the official start date of your research project.
- The period of review shall take between 2 weeks and 1 month, depending on the complexities of your research.
- Do not reach out to individual committee members while you await the outcome of your application.
- Your application will only start going through the review process if you have linked the form to your research proposal and your data collection instruments - with the consent Form on the first page of the instruments;
- You and your supervisor need to sign off on the application with electronic signatures.

PART A: Summary (Box A7 is restricted to keep this section concise).

A1. Campus/Hub/Organisation:	African Leadership University (ALU)
A2. Faculty/Department	BSc. in Software Engineering
A3. Project Title:	A Machine Learning Early-Warning System for Predicting Village-Level Chronic Childhood Stunting Risk in Rwanda Using Multi-Source Data Fusion
A4. Student/Principal Researcher's Name	Thierry SHYAKA
A5. Email address: (institutional email ONLY)	t.shyakal@alustudent.com

A6. Supervisor's Name:

Dirac MURAIRI

Links to Documents:

1. Research Proposal -
[https://docs.google.com/document/d/16v_vhPrDIUGmjZY8UZIfXjXusgC30hU7/edit?usp=sharing&oid=101303438003707586868&rtpof=true&sd=true]
2. Data Collection Instrument - [Not Applicable N/A]
3. Consent Form - customised - [Not Applicable N/A] - [template](#)

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A7. Project Summary: What are you intending to do? Be as explicit as possible about the rationale behind your proposed study,

1. its aims,
2. the methods to be used,
3. including details of your intended procedure and tasks,
4. participant sample and recruitment method,
5. and the location in which the research will take place.

For Software Engineering projects, briefly describe:

- a) Clearly state the nature of the software being developed (e.g., full-stack web application, mobile application, machine learning model, AR/VR simulation, low-level system utility, or a combination of these specialties).
- b) data sources, algorithm, or code to be reused
- c) Provide a high-level overview of how users are expected to interact with the software and the types of data that might be collected, even if detailed plans will emerge later.

Respond to A7 here

PROJECT SUMMARY

1. Aims

My system aims to reduce childhood stunting in Rwanda by shifting the country's nutrition response from reactive screening to predictive prevention. Specifically, the project develops a machine learning early-warning system that forecasts village-level chronic stunting risk three to six months in advance, enabling Community Health Workers (CHWs) and district nutrition officers to intervene during the critical first 1,000 days, before the damage becomes irreversible. The rationale is that Rwanda already has the policy framework (Vision 2050), the financing (World Bank SPRP), and the infrastructure (45,000 CHWs across 14,837 villages); what is missing is a predictive targeting layer that directs limited resources to villages where risk is rising.

2. Methods

The project uses a mixed-methods approach. The primary quantitative method is supervised machine learning: a LightGBM gradient-boosted tree model trained on a longitudinal village-month dataset (~888,000 rows, ~80–100 engineered features), with SHAP for explainability. The secondary qualitative method is semi-structured interviews with CHWs and district nutrition officers to validate the usability of the dashboard, chatbot, and SMS channels. The model is benchmarked against a DHS-only baseline to quantify the marginal value of multi-source fusion.

3. Procedure and Tasks

Implementation follows a 9-week Agile-incremental plan (May 21 – July 27, 2026): Weeks 1–2 acquire and integrate data from DHS, RAB, eSoko, CHIRPS, and MODIS NDVI; Week 3 delivers an end-to-end baseline; Weeks 4–5 refine the model with SHAP and build the dashboard; Week 6 integrates SMS alerts via Africa's Talking; Week 7 builds the Kinyarwanda chatbot via Gemini + RAG over MoH/UNICEF guidelines; Week 8 conducts integration testing and field validation; Week 9 finalises the report and defense.

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4. Participant Sample and Recruitment

The machine learning component uses secondary, anonymized, publicly available aggregate data with **no human participants**. For the qualitative usability validation, a purposive sample of 8–12 CHWs and 2–3 district nutrition officers will be recruited from 2–3 pilot districts. Recruitment proceeds through formal letters to District Health Offices, with participation voluntary, informed consent obtained in writing, and no personally identifiable information beyond signed consent records is retained from interview participants.

5. Location

The research is conducted at African Leadership University (ALU), Kigali, Rwanda. The model covers all 14,837 villages in Rwanda. Field validation interviews take place in 2–3 pilot districts, selected in consultation with district health offices from among the 13 highest-stunting-burden districts identified by the World Bank Stunting Prevention and Reduction Project.

SOFTWARE ENGINEERING DETAILS

1. Nature of the Software

My system is a **full-stack web application combined with a machine learning model and an SMS service**, a hybrid system comprising a supervised ML model (LightGBM with SHAP explainability), a Next.js + TypeScript web dashboard with Leaflet geospatial visualization, a FastAPI Python backend, and dual-channel last-mile delivery (a Kinyarwanda/English RAG chatbot via Gemini and SMS alerts via Africa's Talking). The system is deployed on Render with a PostgreSQL database. The dual-channel delivery layer is what makes Kurinda accessible to both smartphone-using CHW supervisors and feature-phone-using rural CHWs — a core differentiator of the project.

2. Data Sources, Algorithms, and Reused Code

All data sources are public: Rwanda DHS 2024–25 (NISR), Rwanda Agriculture Board crop production data, eSoko market prices, CHIRPS satellite rainfall and MODIS NDVI vegetation indices via Google Earth Engine, and NISR administrative shapefiles. Reused open-source components include LightGBM, scikit-learn, SHAP, GeoPandas, FastAPI, Next.js, Leaflet, Tailwind CSS, shadcn/ui, LangChain, ChromaDB, and the Google Gemini API. Reused knowledge sources for the chatbot RAG layer include Ministry of Health and UNICEF nutrition guidelines (cited and attributed). No proprietary code is reused; the predictive model and the integration architecture are original to this project.

3. User Interaction and Data Collected

Three user groups interact with my system through three channels. District nutrition officers use the web dashboard to view a color-coded village-level risk heatmap, drill down into a village's predicted risk score and SHAP-based explanation, and export action reports. CHW supervisors receive a prioritized village list and may query a Kinyarwanda chatbot (e.g., "Which villages in my sector are highest risk this week?"). Community Health Workers in rural areas receive automated Kinyarwanda SMS alerts when their village's risk score crosses a threshold and may reply by SMS to confirm a home visit.

Data collected by the system is **operational, not clinical**: model predictions (village-level risk scores and SHAP drivers), user account records (name, role, district, sector, phone number, language preference), intervention logs (which village received which intervention and when), and SMS delivery/reply metadata. **No individual patient**

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records, named children, or personally identifiable health information are collected or stored. Operational data (CHW phone numbers for SMS delivery) is retained only during the pilot and securely deleted at project close. All raw input data used for model training is aggregated, anonymized, and publicly sourced.

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Part B: Assessment of Risk

Does your proposed research study involve any of the following? (Click the appropriate answer box. Striking it out means that it is your choice of the options)

B1. My Research involves vulnerable groups (such as children aged 16 and under, those lacking capacity, or individuals in a dependent or unequal relationship)	☐ YES ☒ NO
B2. Research involves sensitive topics (such as participants' sexual behaviour; their legal or political behaviour; their experience of violence; their gender or ethnic status)	☐ YES ☒ NO
B3. Research involves a significant element of deception (i.e., beyond the withholding of the research hypotheses, where this is necessary for methodological reasons)	☐ YES ☒ NO
B4. Research involves access to records of personal or confidential information (including genetic or other biological information)	☐ YES ☒ NO
B5. Research involves access to potentially sensitive data through third parties (such as organisational or employee data)	☐ YES ☒ NO
B5.1 Does research involve collecting, storing, or processing Personally Identifiable Information (PII) or other sensitive user data (e.g., health, financial, behavioural)?	☐ YES ☒ NO
B5.2 Research project utilises third-party services (e.g., cloud hosting, databases, analytics platforms, APIs)	☒ YES ☐ NO
B6. Research that could induce psychological stress, anxiety, or humiliation, or cause more than minimal pain (e.g., repetitive or prolonged testing)	☐ YES ☒ NO
B7. Research that may place the researcher at risk of psychological or physical harm	☐ YES ☒ NO
B8. Research that is conducted off-campus (e.g., in an archive or library, at a community centre, at a constituency office, during filming, photography, recording, or interviewing, etc.)	☐ YES ☒ NO
B9. Research involving invasive interventions (such as the administration of drugs or other substances, vigorous physical exercise, or techniques such as hypnotherapy) that would not usually be encountered during everyday life	☐ YES ☒ NO
B10. Research that may have an adverse impact on employment or social standing (e.g., discussion of an employer, discussion of commercially sensitive information)	☐ YES ☒ NO
B11. Research that may lead to 'labelling' either by the researcher (e.g., categorisation) or by the participant (e.g., 'I am stupid', 'I am not normal')	☐ YES ☒ NO
B12. Research that involves the collection of human tissue, blood, or other biological samples	☐ YES ☒ NO
B13. Research using procedures that may interact with a pre-existing medical condition in a participant (e.g., a heart disorder in physical exercise studies, allergies in taste studies, epilepsy in computer-based studies).	☐ YES ☒ NO
B14. Research requiring the use of potentially hazardous equipment or environments.	☐ YES ☒ NO
B15. Research requiring ethical approval from another source (e.g., research in prisons).	☐ YES ☒ NO
B16. Research requiring permissions from another source (e.g., schools or businesses/organisations, a copyright holder, commercial sources).	☐ YES ☒ NO
B17. Research involves the development or use of software that collects or processes personal data	☐ YES ☒ NO
B18. Research involves the use of artificial intelligence/machine learning models.	☒ YES ☐ NO

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B19. Research uses external datasets	☒ YES ☐ NO
B20. Research involves reusing external code or datasets.	☐ YES ☒ NO
B21. Research involves human-computer interaction (e.g., user testing).	☐ YES ☒ NO

If you answered 'Yes' to any of the above (Section B), please explain how you will ensure the ethical conduct of the research.

If you answered 'Yes' to B17, B18, B19, B20, and B21, give explanations (e.g., for how data is secured, measures to address bias, licenses and attribution, consent process).

If applicable, explain what steps you will be taking to ensure you are appropriately skilled to collect data from special populations or to collect data on sensitive issues. You must demonstrate here that you have anticipated and planned for any difficulties that may arise, including consideration of any personal risk that the researcher may be exposed to (including working off-site).

If you answered "yes" to any question in section B "Assessment of risk," then respond here - in this box.

Indicate the number (e.g., B16), then explain how you will ensure ethical conduct as per the specific question:

Example:

B16 - ethical conduct will be maintained by...

ETHICAL CONDUCT

B5.2: Use of Third-Party Services (cloud hosting, databases, APIs)

Ethical conduct will be maintained by selecting only reputable third-party services with documented data protection commitments: Render for hosting (frontend and backend), PostgreSQL for the database, Google Cloud (Earth Engine, Gemini API) for satellite data and the chatbot LLM, and Africa's Talking for SMS delivery. All services will be used within the scope of their published terms of service and privacy policies. No personally identifiable health information will be transmitted to any external service; only operational data necessary for system function (e.g., phone numbers for SMS delivery, model predictions, anonymized aggregate inputs) will be processed externally. API credentials and secrets will be stored in environment variables, never committed to version control. All data in transit between users, the backend API, the database, and third-party services is protected by HTTPS/TLS encryption, and PostgreSQL connections use SSL. Where free tiers are used, data residency and retention policies will be reviewed before any pilot data is uploaded, and CHW phone numbers will be securely deleted from Africa's Talking at project close.

B18: Use of Artificial Intelligence / Machine Learning Models

Ethical conduct will be maintained by adhering to the principles of transparency, fairness, accountability, and non-replacement of clinical judgment. Every model prediction will be paired with SHAP-based plain-language explanations of its top three risk drivers, enabling end-users to interrogate and contest model outputs rather than treat them as black-box decisions. Model performance will be evaluated for fairness across rural/urban and high-/low-resource districts to detect and mitigate disparate impact, and these results will

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be reported transparently in the final report. Known sampling biases in DHS data — including under-representation of mobile populations, remote areas, and informal settlements will be documented in the final report and accounted for during model interpretation; predictions for under-sampled regions will be flagged with reduced confidence. The chatbot LLM (Google Gemini) will be constrained via Retrieval-Augmented Generation to answer only from verified Ministry of Health and UNICEF nutrition guidelines and from my system's own risk data, with explicit refusal of out-of-scope queries to prevent hallucinated medical advice. Chatbot queries sent to the Gemini API will use the production tier with data-not-used-for-training guarantees, ensuring no CHW or officer query becomes training data for future models; the free tier may be used during early development with synthetic test queries only. The system will be framed in its user interface as a decision-support and prioritisation tool, not a substitute for in-person clinical assessment by a Community Health Worker, and all predictions will be reviewed by qualified district nutrition officers before any intervention is dispatched.

B19: Use of External Datasets

Ethical conduct will be maintained by using only datasets that are public, anonymized, and aggregated, and by complying with each provider's data use agreement. The Rwanda Demographic and Health Survey (DHS) microdata will be accessed only after formal registration and acceptance of the DHS Program's data use terms, which restrict use to research purposes and prohibit re-identification attempts. Rwanda Agriculture Board, NISR, eSoko, CHIRPS, and MODIS NDVI datasets are publicly published and licensed for research and educational use; all sources will be properly cited in the final report and code repository. No individual-level patient records, named children, or personally identifiable health information will be downloaded, processed, or stored at any stage. Project-derived datasets (the engineered village-month feature matrix and the trained model artefacts) will be retained only for the duration of the project and securely deleted at project close. Raw public datasets may be re-downloaded from their original sources at any time and therefore do not require local retention beyond the active development window.

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PART C: Data Management and Storage

During Data Collection

All paper forms involving personal data (such as consent forms) should immediately be scanned, and the electronic files should be stored on your personal Google Drive at African Leadership University as an electronic backup. For coding situations, resort to private, university-sanctioned version control repositories (e.g., private GitHub or GitLab instances) with appropriate access controls for storing and managing source code. You should not give permission for anyone else to view these files other than your supervisor (who is part of the research team). Under no circumstances should data be stored on unencrypted USB memory sticks or hard drives. In the event of the hardware failing or theft, this means that you will lose the data (which means that people have given their time for nothing, which is unethical). Participants should be allocated a Unique Identifier (ID) code prior to any data collection, and only this ID (not personal information such as name, address – as you have been taught in your Research Symposium Classes) should be recorded on any data collection forms (e.g., interviews, questionnaires, computer tasks). If qualitative data such as focus group recordings are going to be transcribed, then this should be done as soon as possible, then the original recording deleted (upon supervisor confirmation) and the transcriptions pseudo-anonymised to remove personal identifying information (names) – especially for sensitive research. If they are not going to be transcribed, then data reduction and analysis should be done as soon as possible, pseudo-anonymised, and then the recordings should be destroyed.

Data Storage and Retention

All digital materials relating to this study, including data, recordings, and signed consent forms, will be stored on your personal Google Drive at African Leadership University for a period of 3 months following your graduation and will then be securely destroyed.

PART D: Final checklist and declaration

D1. In completing this form, have you followed the guidance provided by your subject area? ☒ YES ☐ NO
D2. Have you included any adverts or letters by which you mean to recruit participants? ☐ YES ☐ NO ☒ NOT APPLICABLE
D3. Will the timescales within which participants may withdraw from the study (including their data) be clear and specific, and explained to participants? ☐ YES ☐ NO ☒ NOT APPLICABLE
D4. Have you made the mechanisms for withdrawing from the study clear to participants, including specifying a point after which it will not be possible to withdraw data? ☐ YES ☐ NO

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☒ NOT APPLICABLE
D5. Have you included your Consent Form? (Please note: Consent Forms are not required for purely postal or internet-based questionnaire studies where completion of the questionnaire itself denotes participant consent). ☐ YES ☐ NO ☒ NOT APPLICABLE
D6. Have you made it clear to participants that the research is being undertaken with supervision for an undergraduate project at African Leadership University? ☐ YES ☐ NO ☒ NOT APPLICABLE
D7. You will provide participants with your supervisor's name and contact details - in case they need to reach out for confirmation or have any future queries? ☐ YES ☐ NO ☒ NOT APPLICABLE
D8. You will provide participants with your name and contact details (i.e., your <u>University</u> email address)? ☐ YES ☐ NO ☒ NOT APPLICABLE
D9. Have you included copies of all measures, tests, inventories, questionnaires, interview questions, and research proposal you intend to use in the course of your research in your application? ☐ YES ☐ NO ☒ NOT APPLICABLE
D10. Have you included approval obtained from other Research Ethics Committees or permissions from other organisations (if applicable)? ☐ YES ☐ NO ☒ NOT APPLICABLE
D11. Have you made it clear to participants that any personal or sensitive data and related materials will be anonymised and stored securely at ALU for a period of 1 year after completion of the study? ☐ YES ☐ NO ☒ NOT APPLICABLE
D12. Will you make it clear who will have access to the raw and summarised data? ☐ YES ☒ NO
D13. Will you make it clear to participants that anonymised data may form part of a publication written by the researcher and/or their supervisor?

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☐ YES ☐ NO ☒ NOT APPLICABLE
D14. Have you checked all participant materials to ensure that there are no spelling or grammatical errors? ☐ YES ☐ NO ☒ NOT APPLICABLE
D15. I confirm that participants will not be subjected to any undue or disproportionate incentives to participate in the proposed study. ☐ YES ☐ NO ☒ NOT APPLICABLE
D16. Have you identified all third-party software, libraries, APIs, or Open-Source Software to be used, verified their licenses, and outlined how you will ensure compliance with all license terms ☒ YES — ☐ NO ☐ NOT APPLICABLE
D17. For software projects: Have you included a plan for secure handling of user data (if any) during testing? ☐ YES ☒ NO — ☐ NOT APPLICABLE
D18. For software projects: If using external code, have you checked the licenses and provided proper attribution? ☐ YES ☐ NO ☒ NOT APPLICABLE
D19. For software projects: If using ML models and external datasets, have you considered potential biases, outlined steps to mitigate such bias, and considered the fairness and accountability of your model, and documented them? ☒ YES — ☐ NO ☐ NOT APPLICABLE

Note that in submitting this form, you are agreeing to comply with the following procedures. Failure to adhere to this process will constitute a breach of ethical requirements and may result in your failing the module or facing disciplinary action:

1. Only you and your supervisor(s) may have access to the data.
2. Prior to commencing data analysis, you must show your supervisor that ALL of your collected data is stored electronically on Google Drive (e.g., questionnaires, response sheets, recordings of interviews, or, for computer-based studies, all of the individual participant data files) so that it can be verified that the data have been collected.
3. Once data analysis has been completed, you must give your supervisor all of the collected data, as described in 2. This will be stored securely and retained for 3

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months after graduation. Any other copies of data that the student/lead investigator(s) may have MUST be destroyed securely once the report has been submitted.

4. Each Faculty must appoint a member of academic staff to the role of Data Management for undergraduate projects, with responsibility for secure storage and destruction of data 3 months to a year after graduation

Supervisor Authorisation

Your application for research ethics review and clearance will NOT be reviewed by the research ethics committee if your supervisor has not authorised and signed this form.

TO BE COMPLETED BY THE STUDY SUPERVISOR

I confirm that the student/lead investigator has had appropriate health and safety training.

- ☒ YES —
☐ NO
☐ N/A

I confirm that I have checked the form and the related materials and agree that, in addition to general ethical issues, the matters listed above have been specifically addressed.

- ☒ YES —
☐ NO

Do not submit this form without your supervisor's approval and signature!

Signature of Study Supervisor(s):

Name(s): Dirac Murairi



Signature of Student(s)/lead investigator(s):

Name(s): Thierry SHYAKA

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